Do Now · retrieval

BIOS, drivers and virtual machines

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does an OS driver do?

Q2

Define multi-user OS.

Q3

What executes before the OS is loaded?

Q4

Simplify (¬A∧B)∨(A∧B). Show factoring.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — BIOS/UEFI firmware initialises key hardware and starts the boot process.

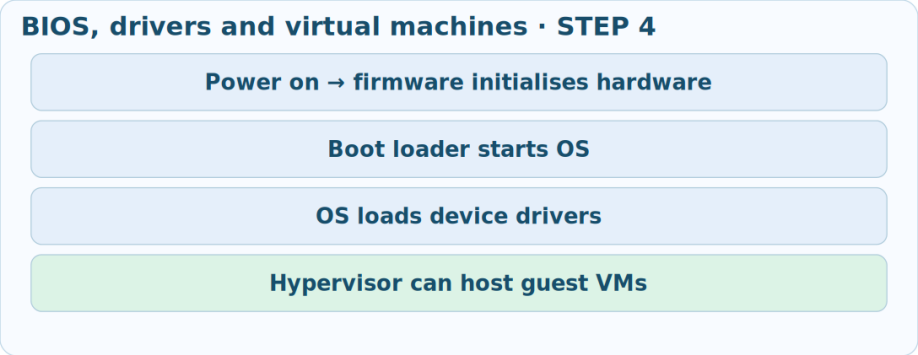
Core idea 2

Trace — Device drivers connect OS abstractions to the instructions and data expected by actual peripherals.

Core idea 3

Evaluate — A hypervisor provides virtual machines for isolation, testing and consolidation, though CPU/RAM/storage overhead remains.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

UEFI — Modern system boot firmware.

driver — Hardware-specific software interface.

VM — Virtual machine.

hypervisor — VM hosting software.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

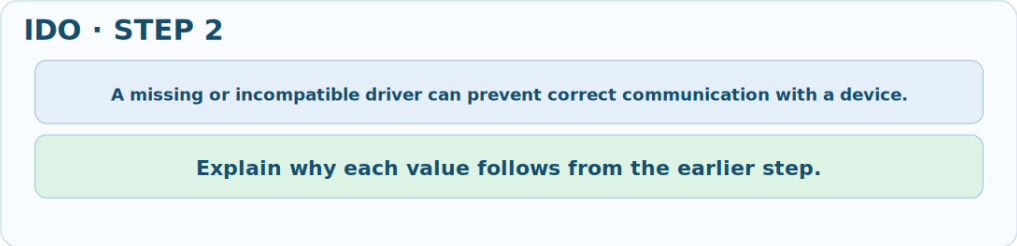
A printer is connected but software cannot print; name an OS dependency.

Follow the worked method

1. Predict first — A missing or incompatible driver can prevent correct communication with a device.

2. Model and check — Explain why each value follows from the earlier step.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
A VM provides a reusable guest environment with snapshots, although it consumes host resources.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

State roles of UEFI/BIOS, boot loader, driver and hypervisor. [4]

Question 2

Explain how a compatible printer driver enables printing. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate guest virtual machines for school testing of unfamiliar operating systems. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A virtual machine has its own wholly separate physical processor.

Use key vocabulary from memory

1. Explain UEFI in your own words.

2. Explain driver in your own words.

3. Explain VM in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.